

Quiz - Lecture 07

Close your laptops and put away all electronics. No electronics or notes are allowed during the duration of this quiz.

Question 1

What is a fruitful function?

- A. A function that returns a result
- B. A function that does not return a result
- C. A function that prints a result
- D. An empty function

Question 2

What keyword is used to send a value back from a function?

- A. `for`
- B. `break`
- C. `print`
- D. `return`

Question 3

What is the output of the following code?

```
Python
def f(x):
    if x > 0:
        return x + 1
    else:
        return x - 1

print(f(f(-1)))
```

- A. It encounters a RecursionError and crashes
- B. -3
- C. -1
- D. 1

Question 4

What is the output of the following code?

```
Python
def g(x):
    if x % 2 == 0:
        return x // 2
    return x * 3 + 1

print(g(g(2)))
```

- A. 1
- B. 2
- C. 3
- D. 4

Question 5

Why are return values useful?

- A. They make programs run faster
- B. They allow results to be reused in expressions
- C. They prevent syntax errors
- D. They replace parameters

Question 6

What happens if a function has multiple `return` statements?

- A. All of them execute
- B. Only the first one is used
- C. Only one is executed per function call
- D. Python raises an error

Question 7

Which of the following can a Python function return?

- A. Only strings
- B. Only booleans
- C. Only integers
- D. Any Python object

Question 8

What happens when a `return` statement is executed?

- A. The program stops entirely
- B. The function exits and sends a value back to the caller
- C. The loop restarts
- D. The value is printed

Question 9

What is the output of this code?

```
Python
def foo():
    return

print(foo())
```

- A. None
- B. 0
- C. An error
- D. False

Question 10

Which of the following best describes the flow of a recursive function?

- A. It calls itself until a base case is reached
- B. It repeats without stopping
- C. It runs line by line once
- D. It executes all calls at once

Bonus Question 1

What is the output of the following code?

Python

```
def test(x):  
    if x > 5:  
        return x + 1  
    if x > 2:  
        return x * 2  
    return x + 10  
  
print(test(test(test(3))))
```

- A. 6
- B. 7
- C. 8
- D. 13

Bonus Question 2

What is the output of the following code?

Python

```
def is_even(n):  
    return n % 2 == 0  
  
def check(x):  
    if is_even(x):  
        return x + 2  
    return x - 1  
  
print(check(check(3)))
```

- A. 3
- B. 4
- C. 5
- D. 6

Bonus Question 3

What does the following function return?

Python

```
def compare(a, b):  
    if a > b:  
        return 1  
    elif a == b:  
        return 0  
    else:  
        return -1
```

- A. Always 1
- B. A boolean value (True or False)
- C. 1, 0, or -1
- D. None